

Fall 2026

CS 415/515

Social Media Data Science Pipelines

Last updated: August 23, 2026

Lecture location: Science Library 310

Lecture time: Monday & Wednesday 5:00pm - 6:30pm

Instructor: Kai-Cheng Yang

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Office hours: Wednesday 2:30pm - 4:30pm, by appointment

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COURSE DESCRIPTION

The focus of this course is on applying data science techniques to large-scale social media. The topics covered include large-scale data collection and management, exploratory analysis and measurement techniques, network analysis, data visualization, hypothesis testing and statistical modeling, and predictive, real-time analytics. Students will build an end-to-end analysis pipeline and use it to answer questions about online events as they occur. The goal of the class is to provide students with a methodological toolbox, the technical skills to make use of these tools, and the experience of using them on real world data.

USEFUL LINKS

The course website is <https://yangkclab.github.io/social-media-ds-course>, which contains the links to the schedule, lectures, and other course resources. You can also find the up-to-date syllabus on the website. Please check the website regularly for updates.

CREDIT HOURS

This course is cross-listed as CS 415 and CS 515.

Both sections are 3 credit hours, which means that in addition to the scheduled lectures/discussions, students are expected to do at least 6.5 hours of course-related work each week during the semester. This includes things like: completing assigned readings, participating in lab sessions, studying for tests and examinations, preparing written assignments, and other tasks that must be completed to earn credit in the course.

COURSE OBJECTIVES

This course is designed to provide a solid foundation and background in performing data science on social media. In particular, upon successful completion of this course, you will be able to:

- Build a continuous data system for social media.
- Manage collected data.
- Design and execute various measurements on social media.
- Model and analyze online behavior via social media.
- Create visualizations that help understand social media phenomena.

PREREQUISITES AND CO-REQUISITES

- CS 350 Operating Systems
- CS 375 Design & Analysis Algorithm
- MATH 327 Probability with Stat Methods or equivalent
- Know at least one programming language well

RELATIONSHIP WITH ABET

- Student Outcome 1 (Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions): This course is a designated course for performance indicator 1-6, “Can understand the principles of information and data management.” Graded work from this course will be used to assess that indicator for the BSCS program.
- Student Outcome 5 (Function effectively as a member or leader of a team engaged in activities appropriate to the program’s discipline): All programming projects are required team projects of 2-3 students.

TEXTBOOK AND REFERENCE BOOKS

Material in this class is delivered via lecture and reading research papers; there is no textbook.

COURSE FORMAT AND TOPICS

This class combines lectures with research paper reading and discussions.

The lectures will cover the fundamentals of data science on social media. The following is a non-exhaustive list of topics that will be covered in the lectures:

- What is Data Science and what does social media have to do with it?
- Data collection
- Social media data formats
- Social media data management with RDBMS/NoSQL
- Applications of probability and statistics, with an emphasis on hypothesis testing
- Network analysis
- Applications of Machine Learning and generative AI
- Visualization

The reading materials will be recent research papers that are related to the topics covered in the lectures. The main topics that will be covered in the reading materials will be:

- Dataset and data collection
- Algorithmic bias
- Inauthentic behaviors
- Ethics and data access
- Generative AI and social media
- Social media of AI agents

LECTURE NOTES AND SUPPLEMENTAL MATERIALS

- Lecture notes will be provided via PDFs or other formats delivered in class.
- All paper reading assignments are listed on the course schedule, which is linked from the course website.

ASSIGNMENTS

- Paper readings. The best way to start understanding what you can do with social media data is to read research papers. There will be regular paper readings, and each student is required to write a short reflection on each reading before the class it is discussed in. See the Paper Reading Reflections section below for how they work. There will potentially be in-class quizzes as well. Students are also expected to participate in the discussion in the class.
- There will be three projects. Each project has three parts: a proposal, an implementation, and a report. Projects are to be completed in groups of 2-3 students.

- Midterm exam. There will be one in-class midterm exam on data collection and data management. See the Midterm Exam section below.
- Final demonstration. Students will be required to demonstrate their project to the class at the end of the semester.

Important notes about assignments:

- Late assignments may sometimes be accepted with penalty, which will typically be 5% per day late (including weekends and holidays). We will not accept assignments more than 5 days after the due date unless there is a very compelling reason.
- For project submissions, please ensure your programs are clear, well-structured, and easy to evaluate. All programming assignments should have:
 - An adequate explanation of the design of your program.
 - * You should be prepared to answer good-faith, technical questions asked about your design and implementation during 1:1 sessions with the instructor.
 - Documented code:
 - * Ideally, you use whatever documentation tools are available in the language you decide to implement in, but at minimum, all modules, classes, and functions should have a documentation header that explains what the code does.
- Grading disputes, regrading and missing grades.
 - Should you dispute any grading, please be aware that we will not re-grade any single issue you have. Instead, your work will be re-evaluated from scratch. The new grade may be higher, lower, or stay the same. This new grade will not be changed.
 - No regrading can be requested two weeks after the date when graded work is returned to students.
 - The scores of your assignments will be made available to you after the assignments are graded.

PAPER READING REFLECTIONS

Reading research papers is how you learn what social media data can be used for. The reflections are how you prepare for the in-class discussion of each paper.

- **How many.** Expect one to two papers per lecture, and roughly 20 papers over the semester. The reading list is on the course schedule, which is linked from the course website.
- **When it is due.** Before the lecture that covers the paper. The reflections exist to prepare you for the discussion in that class, so a reflection added afterwards does not serve its purpose.
- **How long.** A reflection does not need to be long, but it does need to be substantive. Restating the title or the abstract is not a reflection.
- **What to write about.** Anything substantive about the reading. For example: the main idea of the paper, why you like it or do not, what its limitations are, or what future work it suggests.

Reflections and in-class quizzes together are worth 15% of your grade.

MIDTERM EXAM

There is one midterm exam. It is held in class on Monday, October 5th, 2026.

- **Individual and closed-book.** No notes, no books, no electronic devices, and no generative AI tools.
- **What it covers.** The data collection and data management unit, which runs from the start of the semester through September 30th. That is collecting data from social media platforms, data formats, and databases.
- **What it asks for.** The exam tests whether you understand the principles behind that material. For example, the questions might ask you to reason about how to design a collection system, how to choose a data format or a database schema, and what the trade-offs of a storage decision are. You will not be asked to recall the name of a specific API endpoint.

The midterm is worth 15% of your grade.

METHOD OF ASSESSMENT

The following percentage weights will be used to assess student work:

- Paper reading reflections and quizzes are worth 15%.
- The midterm exam is worth 15%.
- Three programming assignments (projects) are worth 60% split evenly across all three projects.
- Final demonstration is worth 10%.

GRADING DETERMINATION

Your final grade for this course is largely based on your performance relative to the performance of other students in the class. In other words, if your work is consistently better than average, you are likely to receive an A. The specific break down of grades is:

- A: 100–90
- B: 89–80
- C: 79–70
- D: 69–60
- F: 59–0

There are no +/- grades.

ACADEMIC HONESTY EXPECTATIONS AND VIOLATION PENALTY

- Cheating on quizzes of any kind, including, but not limited to, the use of electronic devices, “cheat sheets,” or looking at another student’s quiz are considered instances of cheating and will be reported as Category 1 academic dishonesty violation. You will also receive a one letter grade deduction (e.g., from A to B). More than one incident of cheating of any kind will result in an F for the entire course.

- The School of Computing at Binghamton wrote a letter to all computer science students about the importance of academic honesty. The letter is available at <https://www.binghamton.edu/watson/about/academic-honesty.html>.
- The Watson College academic honesty policy defines the violation categories and the procedure that follows a report. The policy is available at <https://www.binghamton.edu/watson/about/honesty-policy.pdf>.
- Please review the academic honesty document and make sure that you understand it!
- Each assignment must include the following statement, verbatim, followed by your group members' names in a file called "HONESTY.md":

“We have done this assignment completely on our own. We have not copied it, nor have we given our solution to anyone else. We understand that if we are involved in plagiarism or cheating, we will have to sign an official form that we have cheated and that this form will be stored in our official university records. We also understand that we will receive a grade of 0 for the involved assignment and our grade will be reduced by at least one level (e.g., from A to B) for our offense, and that we will receive a grade of “F” for the course for any additional offense of any kind.”

Failure to submit your HONESTY.md file with the above text, verbatim, will result in your project not being graded and you receiving a 0 for the submission.

- For this course, programming assignments (projects) are all team projects. Certain open-source tools/software are permitted to be used (see the description of each project for details). Used open-source tools/software must be clearly acknowledged in the submitted project report.
- Additionally, each project submission must include a statement of contribution, which describes which group members did what part of the assignment in a file called "CREDITS.md". Failure to submit your CREDITS.md file will result in your submission not being graded and receiving a 0 for the submission.
- Each project submission must also include an AI usage statement in a file called "AI_USE.md". The Generative AI policy section below describes what it has to contain. It is required for every submission, including from groups that used no generative AI tools. Failure to submit your AI_USE.md file will result in your submission not being graded and receiving a 0 for the submission.
- Stub files for HONESTY.md, CREDITS.md, and AI_USE.md are provided in your project repository. An unedited stub counts as a missing file.
- The use of generative AI tools is allowed for this course given that students follow the guidelines detailed below. Violation of the guidelines and inappropriate use of generative AI tools will be considered cheating and will be reported as Category 1 academic dishonesty violation. More than one incident of cheating of any kind will result in an F for the entire course.

GENERATIVE AI POLICY

Use of generative AI tools is allowed for this course but not encouraged. To maximize learning outcomes, students are encouraged to complete coursework independently. However, we recognize the potential benefits of generative AI tools in the learning process. Below are guidelines for students who choose to use these tools.

Generative AI tools include but are not limited to:

- LLMs, such as ChatGPT, Claude, and Gemini
- Coding assistants, such as Cursor, GitHub Copilot, and Claude Code.

Students are allowed to use generative AI tools to:

- Enhance their understanding of course material, such as gathering information and explaining concepts.
- Clarify ideas and polish their writing.
- Assist with project implementation.
- Process data (e.g., classifying social media posts).

Students should be aware that generative AI outputs can be erroneous, and they are fully responsible for verifying accuracy. Additionally, AI outputs may not meet assignment requirements. Since students are ultimately accountable for their submitted work, the quality will be reflected in their grade.

Every submitted assignment must include an AI usage statement, in a file called "AI_USE.md". For each generative AI tool used, the statement must include the following information:

- The generative AI tool used.
- The prompt used to generate the output.
- How the AI-generated content was integrated into the assignment.
- What the student did to verify that the output was correct.

If no generative AI tools were used for an assignment, the statement must say so explicitly.

The following are not allowed:

- Using generative AI tools to generate entire assignments without modification. Submitted work should reflect the student's own understanding, ideas, and effort.
- Any use of generative AI tools without properly acknowledging it in the AI usage statement.

MANAGING STRESS

If you are experiencing undue personal or academic stress at any time during the semester or need to talk with someone about a personal problem or situation, I encourage you to seek support as soon as possible. I am available to talk with you about stresses related to your work in my class. Additionally, I can assist you in reaching out to any one of a wide range of campus resources, including:

- Dean of Students Office: 607-777-2804
- Decker Student Health Services Center: 607-777-2221
- University Police: On campus emergency, 911 or 607-777-2222
- University Counseling Center: 607-777-2772
- Interpersonal Violence Advocate: 607-777-2804
- Harpur Advising: 607-777-6305
- Office of International Student & Scholar Services: 607-777-2510
- Ombudsman: 607-777-2388
- Services for Students with Disabilities: 607-777-2686 (Voice, TTY)

TITLE IX POLICY

In the event that you choose to write or speak about experiencing or surviving sexual violence (including sexual harassment, dating, and domestic violence), sexual assault, stalking, and rape, please keep in mind that federal and state laws require that, as your instructor, I notify the Title IX Coordinator. He will contact you and provide you with on and off campus resources and discuss your options with you. If you would like to disclose your experience confidentially, you can contact the University Counseling Center, Decker Student Health Services, Harpur's Ferry, Ombudsman, or the Binghamton University Interfaith Council (BUIC).

For more information, please go to the VARCC website (<https://www.binghamton.edu/centers/varcc/index.html>), or the Title IX at Binghamton University website (<https://www.binghamton.edu/services/title-ix/index.html>).

MENTAL HEALTH

Diminished mental health, including significant stress, mood changes, excessive worry, or problems with eating and/or sleeping can interfere with optimal academic performance. The source of symptoms might be largely related to your coursework; if so, I invite you to speak with me (or your other faculty) directly. However, problems with relationships, family worries, loss, or a personal struggle or crisis can also contribute to decreased academic performance and may require additional professional support. Binghamton University provides a variety of support resources: the Dean of Students Office and University Counseling Center offer coaching on ways to reduce the impact on your grades. Both of these resources can help you manage personal challenges that impact your well-being or ability to thrive at Binghamton University. Accessing them, especially early on, as symptoms develop, can help support your academic success as a University student.

In the event I think you could benefit from such support, I will express my concerns (and the reasons for them) to you and remind you of our resources. While I do not need to know the details of what is going on for you, your ability to share some of your situations with me will help me connect you with the appropriate support.

CLASS ATTENDANCE REQUIREMENT

Attendance is required and attendance will be checked regularly. If you are unable to attend class, you **must** notify the instructor **in advance** with a valid reason. Failing to do so will result in a penalty to your final grade. Showing up late is considered missing the class.

COVID-19 POLICY

Students must follow the university's current COVID-19 safety requirements at all times. The full up-to-date guidance is available at <https://www.binghamton.edu/health/services/covid/index.html>.

COMMUNICATION

The course uses Slack for communication. All students are required to join the Slack workspace for the course and check it regularly.

Students can also use email for communication but are required to use their Binghamton email addresses. Emails from non-binghamton.edu domains will not receive a response. All emails must include “[CS415]” or “[CS515]” in the subject line for proper identification. Emails without this subject line identifier may be ignored.

Brightspace will be used for official course announcements and grading.

The instructors will make every effort to respond to student emails within two business days. Please plan accordingly—emails sent shortly before deadlines may not receive timely responses.